

SE-Introduction to programming session-6

(Q.4) Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

Ans. Difference Between Entry-Controlled and Exit-Controlled Loops

Entry-Controlled Loop:-

1. The condition is checked **before** the loop body executes.
2. If the condition is false at the beginning, the loop **does not run even once**.
3. Examples: for, while

Exit-Controlled Loop:-

1. The condition is checked **after** the loop body executes.
2. If the condition is false at the beginning, the loop **runs at least one time**.
3. Example: do...while

What happens if the loop condition is false at the start?

- In an **entry-controlled loop**, if the condition is false at the start, the loop does not execute.
- In an **exit-controlled loop**, if the condition is false at the start, the loop executes once and then stops.